BIOSTAT M235 – Final Project

Does Health Insurance Affect Whether U.S. Adults Receive a
Blood-Sugar Screening? A Causal Analysis of NHANES 2017–2020

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1 Paper Explanation

1.1 Introduction

More than 38 million Americans have diabetes, with an estimated 90–95% of cases being type 2 diabetes [5]. In addition, over one third of American adults have prediabetes, and many of those will progress to type 2 diabetes within three to five years [5]. Because early detection depends on a blood-sugar screening test, understanding what determines whether an individual is screened is an important public-health question.

Health insurance in the United States has been established as a critical determinant of health, including diabetes-related outcomes. Insurance coverage facilitates early diagnosis through access to screening, whereas uninsured individuals may face greater difficulty accessing care, leading to delays in diagnosis and reduced treatment adherence [2]. Prior NHANES analyses reinforce this: insured patients are substantially more likely to meet diabetes care-quality benchmarks, with one study reporting that insured patients had 2.73 times the odds of controlled blood pressure relative to uninsured patients [3]. A further NHANES analysis found that having health insurance was associated with significantly lower odds of undiagnosed prediabetes and type 2 diabetes [9]. The CDC has likewise identified uninsured people as being at elevated risk of undetected diabetes, noting that more than one in four adults with diabetes do not know they have it, and that eight in ten Americans with prediabetes are unaware of their risk [4].

While Mahoney et al. [9] demonstrated that insurance is associated with lower odds of *undiagnosed* diabetes, their outcome was undiagnosed disease status. Our analysis instead targets the step *before* diagnosis: whether an individual receives a blood-sugar screening test at all. We frame this as a causal question and estimate the average treatment effect (ATE) of insurance coverage on screening using three complementary estimators: survey-weighted logistic regression, inverse-probability weighting (IPW), and a doubly robust (AIPW) estimator. We then run a battery of sensitivity analyses.

1.2 Set Up

Data source. We use the National Health and Nutrition Examination Survey (NHANES) 2017–March 2020 pre-pandemic cycle. The exposure is the self-reported binary indicator `HIQ011` ("Are you covered by health insurance or some other kind of health-care plan?"), coded as insured versus uninsured. The outcome is `DIQ180`, self-reported receipt of a blood-sugar screening test in the past three years. Because NHANES skips `DIQ180` for respondents already diagnosed with diabetes, the analytic population consists of *undiagnosed* adults by construction.

Causal estimand. Let A denote insurance coverage ($1 =$ insured, $0 =$ uninsured) and Y the screening outcome. We target the average treatment effect on the risk-difference scale,

$$\tau \;=\; \mathbb{E}\big[Y^1\big] - \mathbb{E}\big[Y^0\big]\,, \tag{1}$$

the change in the population probability of screening that would result if every adult were insured versus uninsured. Identification of τ from observational data rests on three assumptions [7]:

1. **Consistency:** the observed outcome equals the potential outcome under the observed exposure.
2. **Conditional exchangeability (no unmeasured confounding):** $Y^a \perp A \mid L$ for the measured covariate set L.
3. **Positivity:** $0 < \Pr(A = a \mid L = l) < 1$ for every covariate stratum l in the population.

Confounders. The covariate set L was selected using a directed acyclic graph (DAG) constructed from prior literature (see the project repository, `nhanes_dag.Rmd`), choosing variables that plausibly affect both insurance coverage and the propensity to be screened. Table 1 lists the covariates and their rationale. Notably, variables that lie on the causal path from insurance to screening (e.g. having a usual source of care or the number of health-care visits) were deliberately *excluded* as mediators rather than confounders.

Table 1: Covariates included in the adjustment set L.

Covariate	**Rationale**
Age	Older adults are more likely to be insured (e.g. Medicare eligibility at 65) and are more frequently referred for screening due to higher baseline diabetes risk.
Sex	Men and women may differ in healthcare-seeking behavior and insurance-coverage patterns.
Race/ethnicity	Racial and ethnic minorities have systematically lower insurance rates and differ in diabetes risk and screening likelihood.
Education	Higher education is associated with greater health literacy, employer-sponsored insurance, and engagement with preventive care.
Marital status	Married/partnered individuals are more likely to be covered under a spouse's plan and to have higher healthcare utilization.
Country of birth	Foreign-born individuals face barriers to obtaining insurance and accessing preventive screening.
Interview language	Proxy for acculturation.
Income-to-poverty ratio	Lower income is strongly associated with both lack of coverage and reduced preventive-care utilization.
BMI	Elevated BMI is a clinical risk factor for type 2 diabetes, so providers are more likely to order screening regardless of insurance.
Self-rated health	Individuals rating their health poor/fair may seek care more, but may also be less likely to be insured.
Hypertension	A major comorbidity co-occurring with insulin resistance and metabolic syndrome that itself prompts screening.
Ever smoked $\geq$100 cigarettes	Associated with socioeconomic disadvantage, lower insurance rates, and insulin resistance.
Ever had alcohol	Heavy alcohol use is linked to metabolic dysregulation, affecting glucose metabolism and screening decisions.

Sample and weighting. The analytic sample was restricted to adults aged $\geq$ 20 years who answered both the insurance and screening questions. Respondents who completed the interview but not the physical exam (zero MEC weight) were excluded because BMI is measured in the exam. All estimates incorporate the NHANES complex sampling design, namely primary sampling units (`SDMVPSU`), strata (`SDMVSTRA`), and MEC exam weights (`WTMECPRP`), via the `survey` package [8], so that estimates are nationally representative. Covariate missingness was handled by complete-case analysis in the primary models and probed by multiple imputation in the sensitivity analyses. The full exclusion cascade is summarized in the STROBE diagram in the repository.

1.3 Methodology and Main Analysis

We estimated τ three ways. Triangulating across estimators with different modeling assumptions guards against any single method's misspecification.

1.3.1 Inverse Probability Weighting (IPW)

Inverse-probability-of-treatment weighting models the propensity score $e(L) = \Pr(A = 1 \mid L)$ by logistic regression and reweights individuals by the inverse of their probability of the observed exposure, creating a pseudo-population in which exposure is independent of the measured covariates [10]. Stabilized weights were used to reduce variance and were combined with the NHANES survey weights. Covariate balance before and after weighting was assessed with standardized mean differences (love plot). *[Preliminary: IPW-adjusted ATE =* 0.191*, bootstrap 95% CI* $(0.108, 0.269)$*; full write-up to be completed by A. Millatt.]*

1.3.2 Regression Analysis

A survey-weighted logistic regression model was used as the initial method, since the outcome is binary. Three models were fit with increasing adjustment. *Model 1* was unadjusted. *Model 2* added sociodemographic covariates (age, sex, race/ethnicity, education, income-to-poverty ratio, marital status, country of birth, interview language). *Model 3* additionally adjusted for health status and behavior (BMI, self-rated health, hypertension, smoking, alcohol). The dispersion parameter was near one, supporting a binomial model. Center weights (`WTMECPRP`) were used for national representativeness, and observations with missing covariates were dropped. A Wald test assessed the significance of the insurance term in each model, and the ATE was obtained by g-computation (standardization). Results appear in Table 2.

1.3.3 Doubly Robust Estimation (AIPW)

As the primary analysis we used the augmented inverse-probability-weighted (AIPW) estimator, which is *doubly robust*: it combines a propensity model $e(L)$ with an outcome model $m_a(L) = \mathbb{E}[Y \mid A = a, L]$ and remains consistent for τ if *either* model (but not necessarily both) is correctly specified [11, 1, 6]. For each subject i we form the pseudo-outcomes

$$\hat{\mu}_1^{(i)} = m_1(L_i) + \frac{A_i}{e(L_i)}\big(Y_i - m_1(L_i)\big), \tag{2}$$

$$\hat{\mu}_0^{(i)} = m_0(L_i) + \frac{1 - A_i}{1 - e(L_i)}\big(Y_i - m_0(L_i)\big), \tag{3}$$

whose survey-weighted means estimate $\mathbb{E}[Y^1]$ and $\mathbb{E}[Y^0]$; the ATE is $\hat{\tau} = \overline{\hat{\mu}_1} - \overline{\hat{\mu}_0}$. Both nuisance models were logistic regressions over the covariates in Table 1, fit with the NHANES exam weights. (To avoid numerical instability, the exam weights were rescaled to mean one within the nuisance fits; this leaves the coefficients unchanged but prevents the spurious separation that very large case weights induce.) Design-based confidence intervals for the risk difference, risk ratio, and marginal odds ratio were obtained by the delta method applied to the survey-weighted potential-outcome means, so that the NHANES clustering and stratification are respected.

1.3.4 Main Results

After excluding interview-only respondents, the regression analytic sample was 6,894 adults: 5,694 (82.6%) insured and 1,200 (17.4%) uninsured. The weighted screening rate was already markedly different by group: 57% among insured adults versus 34%

among uninsured adults. The doubly robust analysis, which requires complete data on all twelve covariates, used 5,822 adults and estimated $\mathbb{E}[Y^1] = 0.56$ and $\mathbb{E}[Y^0] = 0.37$.

Table 2: Survey-weighted logistic regression: odds ratio (OR) for insurance and g-computation ATE, by level of adjustment.

Model	**OR**	**95% CI**	***p*-value**	**ATE**
1: Unadjusted	2.52	(2.11, 3.01)	< 0.001	0.225
2: Demographically adjusted	1.95	(1.51, 2.52)	0.0002	0.156
3: Fully adjusted	1.94	N/A	0.108	0.147

In the unadjusted model, insured adults had about 2.5 times the odds of being screened (OR = 2.52, 95% CI 2.11–3.01, $p < 0.001$). Adjusting for sociodemographic covariates attenuated this to OR = 1.95 (95% CI 1.51–2.52, $p = 0.0002$); adding health and behavioral variables barely changed it (OR = 1.94). Within Model 2, age (OR = 1.02 per year, $p = 0.0003$) and education (some college OR = 1.41; college graduate OR = 1.45) were significant predictors of screening. Wald tests confirmed insurance as a significant predictor in Model 1 ($F = 115.5$, $p < 0.001$) and Model 2 ($F = 35.0$, $p = 0.0002$). Because NHANES masks its true PSU identifiers, very few residual degrees of freedom remain once Model 3's covariates are included, rendering its confidence intervals unreliable; we therefore rely on Model 2 for inference and report only Model 3's point estimates and p-values.

The three estimators agree closely (Table 3). The doubly robust risk difference was 0.187 (95% CI 0.124–0.249), corresponding to a marginal OR of about 2.1 and a risk ratio of 1.50. Reassuringly, the AIPW estimate's two ingredients computed alone, the plug-in g-formula (0.162) and IPW (0.182), bracket the AIPW value, the hallmark of a well-behaved doubly robust estimate. All three methods indicate that insurance raises the probability of blood-sugar screening by roughly 15–19 percentage points.

Table 3: Average treatment effect (risk difference) across the three estimators.

Method	**ATE (risk difference)**	**95% CI**
Regression (g-computation, Model 2)	0.156	N/A
Inverse probability weighting	0.191	(0.108, 0.269)
Doubly robust (AIPW)	0.187	(0.124, 0.249)

2 Paper Discussion

2.1 Advantages

Using three estimators with different assumptions strengthens our conclusions: agreement across methods makes the finding robust to any one method's misspecification. Logistic regression is straightforward to implement and interpret, yields odds ratios comparable to prior literature, and uses the full analytic sample. IPW explicitly targets the ATE and separates the design stage (propensity modeling and balance checking) from outcome analysis. The doubly robust estimator offers the strongest theoretical guarantee, requiring only one of the two nuisance models to be correct, and, unlike regression alone, directly estimates a marginal causal estimand on the risk-difference scale. All analyses incorporate the NHANES survey design, so estimates generalize to the U.S. adult population.

2.2 Limitations

Several limitations temper the interpretation. First, both exposure and outcome are self-reported and subject to recall and social-desirability bias. Second, NHANES is cross-sectional, so insurance status and screening are measured contemporaneously, and reverse pathways cannot be fully excluded. Third, positivity is strained at the top of the propensity distribution: adults aged 65+ are nearly universally insured through Medicare ($\approx$ 98%), so there are few comparable uninsured older adults, and effect estimates in that region rely on model extrapolation. Fourth, complete-case analysis assumes data are missing at random; income-to-poverty ratio and alcohol use carry the most missingness. Finally, residual confounding from unmeasured factors (e.g. provider behavior, local health-system characteristics) remains possible, which motivates the formal sensitivity analysis below.

2.3 Comments

Across regression, IPW, and doubly robust estimation, having health insurance is associated with a 15–19 percentage-point increase in the probability of receiving a blood-sugar screening among undiagnosed U.S. adults. This extends Mahoney et al. [9] upstream of diagnosis: insurance appears to shape not only whether diabetes is detected, but whether the screening step that enables detection occurs at all.

3 Extension: Sensitivity Analysis

3.1 Motivation

The causal interpretation of τ depends on assumptions that cannot be fully verified from the data: no unmeasured confounding, positivity, a correctly specified adjustment set, and an appropriate treatment of missing data. As an extension, we quantify how fragile the headline doubly robust estimate is to plausible violations of each. All checks re-use the AIPW estimator as the workhorse and are implemented in `sensitivity/sensitivity_analysis.Rmd`.

3.2 Method

We examined four threats. (1) *Unmeasured confounding* was quantified with the E-value [13], the minimum strength of association (on the risk-ratio scale) that an unmeasured confounder would need with both insurance and screening to explain away the estimate. (2) *Positivity* was probed by re-estimating the ATE after truncating the propensity score to progressively tighter bounds and after trimming the sample to the region of common support. (3) *Adjustment-set sensitivity* compared the primary twelve-covariate set against adding alcohol use, a demographic/SES-only set, and a minimal set. (4) *Missing data* was addressed by multiple imputation by chained equations (five imputations), re-estimating the AIPW within each completed dataset and pooling by Rubin's rules [12].

3.3 Results

The primary AIPW risk difference of 0.187 corresponds to an E-value of **2.37**, with the lower confidence bound's E-value equal to 1.82. That is, an unmeasured confounder would need to be associated with both insurance and screening by a risk ratio of at least 2.37, beyond all measured covariates, to fully explain away the effect, which is a substantial degree of confounding. Positivity restrictions (Table 4) attenuated the estimate only modestly, to 0.15–0.16, with every confidence interval still excluding zero. Varying the adjustment set produced risk differences from 0.150 (minimal) to 0.189 (adding alcohol), all significant. Finally, multiple imputation gave a pooled risk difference of 0.168 (95% CI 0.113–0.223), close to the complete-case value of 0.187, indicating that complete-case analysis did not materially bias the estimate.

Table 4: Sensitivity of the doubly robust risk difference (RD).

Check	**Specification**	**RD**	**95% CI**
Positivity	No trimming	0.187	(0.124, 0.249)
	Truncate PS $[0.05, 0.95]$	0.161	(0.124, 0.197)
	Truncate PS $[0.10, 0.90]$	0.160	(0.131, 0.189)
	Trim to PS $[0.05, 0.95]$	0.150	(0.087, 0.212)
Adjustment set	Primary (12 covariates)	0.187	(0.124, 0.249)
	+ alcohol use	0.189	(0.124, 0.254)
	Demographic/SES only	0.169	(0.103, 0.235)
	Minimal	0.150	(0.090, 0.210)
Missing data	Complete case	0.187	(0.124, 0.249)
	Multiple imputation ($m = 5$)	0.168	(0.113, 0.223)

3.4 Discussion

The headline finding is stable across all four checks: the effect direction and rough magnitude persist under tighter positivity restrictions, alternative adjustment sets, and multiple imputation, and the E-value indicates that only a fairly strong unmeasured confounder could overturn it. The largest single movement comes from restricting positivity, which both attenuates the estimate and narrows it toward the region where insured and uninsured adults are genuinely comparable, consistent with the Medicare-driven overlap concern noted among older adults. We therefore interpret 0.15–0.19 as a robust range for the effect of insurance on blood-sugar screening, with the lower end the more defensible estimate once positivity is enforced.

4 Acknowledgments

Sujit Silas Armstrong assembled and cleaned the analytic dataset (`nhanes_data_prep.Rmd`), implemented the doubly robust (AIPW) estimator (`modeling/doubly_robust.Rmd`), and conducted the sensitivity analyses (`sensitivity/sensitivity_analysis.Rmd`). **Janvi Bharucha** wrote the introduction and dataset description and carried out the survey-weighted regression analysis (`modeling/outcome_regression.Rmd`). **Amanda Millatt** constructed the DAG and identification strategy (`nhanes_dag.Rmd`) and performed the inverse-probability-weighting analysis (`modeling/ipw.Rmd`).

Project repository: m235-causal-inference-final-project.

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